Charge

1. Conservation – isolated system total electric charge
2. Quantization

   smallest unit of charge e → elementary
   ↳ proton or electron

   q ← variable

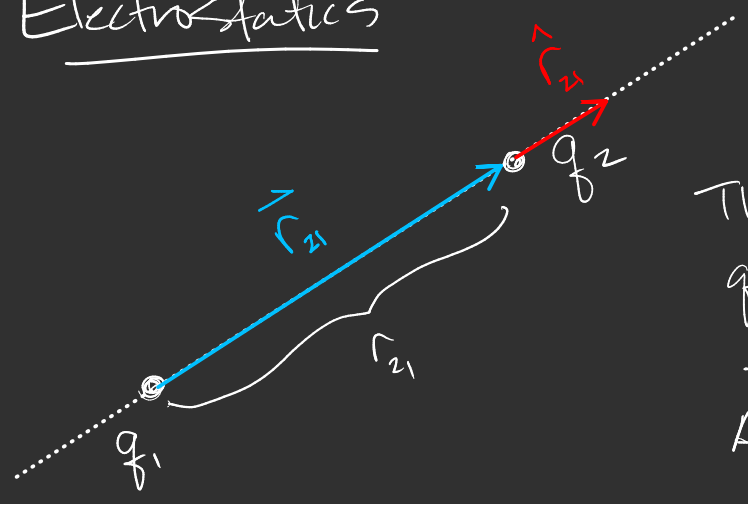
SI units → Coulomb $\quad 1e \equiv 1.602 \cdot 10^{-19}\,C$

$$\downarrow$$

$$1C = \frac{1}{1.602 \cdot 10^{-19}}\,e = 6.242 \cdot 10^{18}\,e$$

Electrostatics

$$\hat{a} = \frac{\vec{a}}{|\vec{a}|} \qquad r^2 = \vec{r} \cdot \vec{r}$$



The force between q_1 + q_2 is directly proportional to the product of q_1 + q_2 AND inversely proportional to distance squared btt them

Coulomb's Law

$$\vec{F}_{21} = \frac{k\, q_1 q_2}{r_{21}^2}\,\hat{r}_{21}$$

Coulomb constant

electric constant

$$k = 8.988 \cdot 10^{9}\ \frac{N\,m^2}{C^2} \approx 9 \cdot 10^{9}\ \frac{N\,m^2}{C^2}$$

$$k = \frac{1}{4\pi\epsilon_0}$$

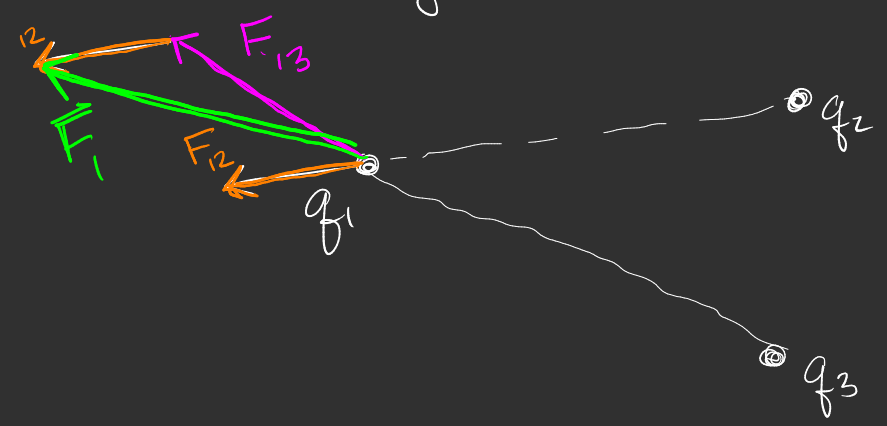
permittivity of free space

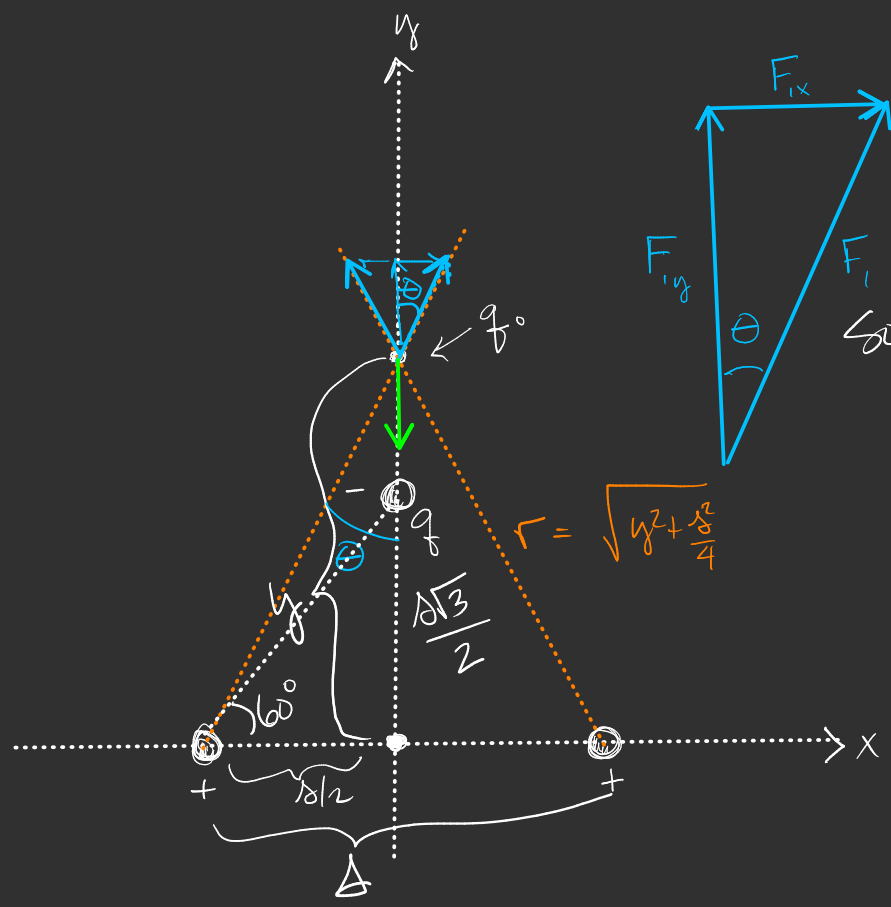
$$\epsilon_0 = 8.85 \cdot 10^{-12}\ \frac{C^2}{N\,m^2}$$

What about more than two charges?

→ Superposition

$$\vec{F}_1 = \vec{F}_{12} + \vec{F}_{13}$$





SOHCAHTOA

$$\cos\theta = \frac{y}{\sqrt{y^2 + \frac{\delta^2}{4}}}$$

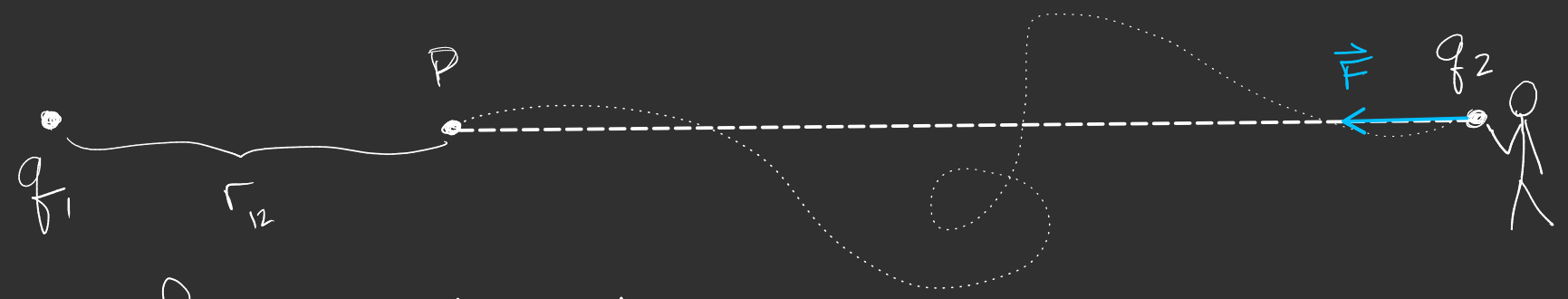
$$0 = F = 2 \cdot \frac{k\cancel{q}q_0}{y^2 + \frac{\delta^2}{4}} \cdot \frac{y}{\sqrt{y^2 + \frac{\delta^2}{4}}} - \frac{k\cancel{q}q_0}{\left(y - \frac{\delta\sqrt{3}}{2}\right)^2}$$

$$\frac{1}{\left(y - \frac{\delta\sqrt{3}}{2}\right)^2} = \frac{2 \cdot y}{\left(y^2 + \frac{\delta^2}{4}\right)^{3/2}}$$

$$y = \frac{\left(y^2 + \frac{\delta^2}{4}\right)^{3/2}}{2\left(y - \frac{\delta\sqrt{3}}{2}\right)^2}$$

$\delta = 1$ $\quad y = 3.1, 0.29, 0.12$?

Work + Energy



$$W = \int \text{force} \cdot \text{displacement}$$

$$W = \int \vec{F} \cdot d\vec{r}$$

Ex. move radially

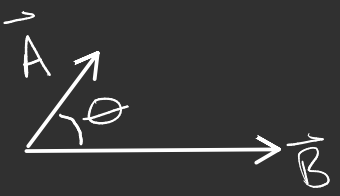
$$W_{ME} = \int_{\infty}^{r_{12}} -\frac{kq_1q_2}{r^2}\,\hat{r} \cdot d\vec{r}$$

$$= -kq_1q_2 \int_{\infty}^{r_{12}} r^{-2}\,dr$$

$$= -kq_1q_2 \left(\frac{(-1)}{r}\right)\Bigg|_{\infty}^{r_{12}}$$

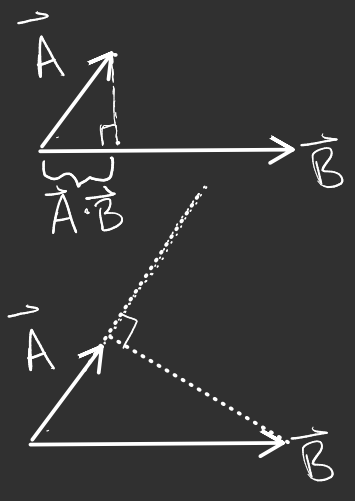
$$\boxed{W_{ME} = \frac{kq_1q_2}{r_{12}}}$$

$$\vec{A} \cdot \vec{B} = |A||B| \cos\theta$$



$$\vec{A} = (A_x\hat{x} + A_y\hat{y} + A_z\hat{z})(B_x\hat{x} + B_y\hat{y} \ldots$$

$$\vec{A} \cdot \vec{B} = A_xB_x + A_yB_y + A_zB_z$$



$$\vec{A} \cdot \vec{B} = \vec{B} \cdot \vec{A}$$